

CS209A 24Fall Java2 Quiz

quiz2

Question 1

1 out of 1 points

Collection is a ____ while Collections is ____

Selected Answer: ☒ Interface, Class

Answers: ☐ Class, Interface

☐ Class, Class

☐ Interface, Interface

☒ Interface, Class

Question 2

1 out of 1 points

What is the primary difference between Comparable and Comparator?

Selected Answer: ☒ Comparable is used to define a natural order, while Comparator allows for custom ordering.

Answers: ☐ Comparable allows for multiple comparison strategies, whereas Comparator allows only one.

☐ Comparable is a functional interface, whereas Comparator is not.

☒ Comparable is used to define a natural order, while Comparator allows for custom ordering.

☐ Comparable is implemented outside the class being compared, while Comparator is implemented within it.

Question 3

1 out of 1 points

Given below code snippet

```
interface A {  
    int aMethod(String s);  
}
```

Which of the following are valid statements?

Selected Answer: ☒ A b = (String s) -> 1;

Answers: ☐ A a = a -> a.length();

☐ A x = y -> {return y};

☐ A s = "2" -> Integer.parseInt(s);

☒ A b = (String s) -> 1;

quiz3

Question 1

10 out of 10 points

What is the value of x?

```
List<String> words = Arrays.asList("Java", "Python", "C++", "JavaScript");  
  
String x = words.stream()  
  
    .reduce((word1, word2) -> word1.length() > word2.length() ? word1 : word2)  
  
    .orElse("");
```

Selected Answer: ☒ JavaScript

Answers: ☐ Java

☐ Python

☐ C++

☒ JavaScript

☐ =

Question 2

10 out of 10 points

Which of the following does Stream map() operates on

Selected Answer: ☒ Function

Answers: ☐ Consumer

☐ Supplier

☐ Predicate

☒ Function

Question 3

10 out of 10 points

Which of the following is terminal operation?

Selected Answer: ☒ max

Answers: ☐ sorted

☐ filter

☒ max

☐ distinct

quiz4

Question 1

1 out of 1 points

Which encoding is known for its variable-length encoding of characters, allowing it to represent a wide range of Unicode characters?

Selected Answer: UTF-8

Answers: UTF-8
GBK
ASCII
ISO-8859-1

Question 2

1 out of 1 points

The size of a Java char variable is _____ bits.

Selected Answer: 16

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
Exact Match	16	

Question 3

1 out of 1 points

What is the primary difference between Reader/Writer classes and InputStream/OutputStream classes?

Selected Answer: Reader/Writer are for reading/writing characters, whereas InputStream/OutputStream are for bytes.

Answers: Reader/Writer are for reading/writing bytes, whereas InputStream/OutputStream are for characters.
Reader/Writer are for reading/writing characters, whereas InputStream/OutputStream are for bytes.
Reader/Writer are faster than InputStream/OutputStream.

quiz5

Question 1

1 out of 1 points

Suppose you have a FileVisitor that needs to keep track of the total number of files visited. Where would you initialize the counter for tracking the number of files?

Selected Answer: Before the traversal starts, outside the visitor methods.

Answers: Inside the visitFile method.
Inside the preVisitDirectory method.
Before the traversal starts, outside the visitor methods.
Inside the postVisitDirectory method.

Question 2

1 out of 1 points

The keyword _____ can be used to prevent a field from being serialized.

Selected Answer: transient

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
Exact Match	transient	

Question 3

1 out of 1 points

What is the main difference between checked and unchecked exceptions?

Selected Answer: Checked exceptions must be thrown in the method signature or caught by try-catch, while unchecked exceptions do not need these.

Answers: Checked exceptions indicate errors that are recoverable, while unchecked exceptions indicate fatal errors that are not recoverable.
Checked exceptions must be thrown in the method signature or caught by try-catch, while unchecked exceptions do not need these.
Checked exceptions are subclasses of Error, while unchecked exceptions are subclasses of Exception.
Checked exceptions are handled at runtime, while unchecked exceptions are handled at compile time.

quiz6

Question 1

10 out of 10 points

Which of the following options is **NOT** the reason for implementing the Runnable interface rather than extending the Thread class, when implementing multithreading?

Selected Answer: Implementing the Runnable interface introduces less concurrency bugs than extending the Thread class.

Answers: Java doesn't support multiple inheritance.
Implementing Runnable helps decouple the task from its execution.
Implementing the Runnable interface introduces less concurrency bugs than extending the Thread class.
Extending a class in OOP typically implies adding new functionality or modifying existing behaviors, which isn't the case when dealing with threads.

Question 2

10 out of 10 points

When a method is declared as synchronized, what does it lock on?

Selected Answer: The object on which the method was called

Answers: The local variables within the method
All methods in the same class
All fields in the same class
The object on which the method was called

Question 3

10 out of 10 points

Which method contains the body of a thread in Java?

Selected Answer: run()

Answers: start()
run()
main()
execute()

quiz7

Question 1

1 out of 1 points

When a thread attempts to acquire a Lock (implemented by the synchronized keyword or the LOCK interface) that is already occupied by another thread, it will enter ____ state.

- Selected Answer:

✔

 BLOCKED
- Answers:

WAITING

✔

 BLOCKED

RUNNING

TERMINATED

Question 2

1 out of 1 points

Which method does the ServerSocket on the server use to receive connections from the client?

- Selected Answer:

✔

 accept()
- Answers:

bind()

listen()

✔

 accept()

receive()

Question 3

0 out of 1 points

In Java's socket programming, a client uses ____ ; a server uses ____

- Selected Answer:

✖

 ClientSocket object: ServerSocket object
- Answers:

ClientSocket object: ServerSocket object

Socket object: ServerSocket object

✔

 Socket object: ServerSocket object and Socket object

Socket object: Socket object

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Question 4

1 out of 1 points

Which interface should a class implement if it is to be executed as a task in a thread pool?

- Selected Answer:

✔

 Runnable or Callable
- Answers:

Task

✔

 Runnable or Callable

Thread

Executor

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quiz8

Question 1

1 out of 1 points

What is the retention policy for @Override?

- Selected Answer:

✔

 SOURCE
- Answers:

✔

 SOURCE

CLASS

BUILDTIME

RUNTIME

Question 2

1 out of 1 points

In Java Reflection, the ____ class is used to represent a class.

- Selected Answer:

✔

 Class
- Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	Class	Case Sensitive

Question 3

1 out of 1 points

What is the main purpose of the Platform.runLater() method in JavaFX?

- Selected Answer:

✔

 To update the UI on the JavaFX Application Thread asynchronously
- Answers:

To run a task in a background thread

✔

 To update the UI on the JavaFX Application Thread asynchronously

To run a task immediately on the main thread

To start a new JavaFX application

Question 4

1 out of 1 points

Which of the following is TRUE about the servlet lifecycle?

- Selected Answer:

✔

 The servlet's destroy() method is called before the container shuts down the servlet.
- Answers:

The servlet container calls init() for each request.

The service() method is called once per servlet lifecycle.

✔

 The servlet's destroy() method is called before the container shuts down the servlet.

The doGet() and doPost() methods are invoked during servlet initialization.

quiz9

Question 1

1 out of 1 points

Which best describes Apache Tomcat?

Selected Answer: ☒ It is a web server and servlet container.

Answers: ☐ It is a full-fledged Java EE application server.

☒ It is a web server and servlet container.

☐ It is used primarily for non-Java web applications.

☐ It is a desktop application development tool.

Question 2

1 out of 1 points

JPA is part of the Java EE specification, while Hibernate is a separate framework that implements JPA.

Selected Answer: ☒ True

Answers: ☒ True
☐ False

Question 3

1 out of 1 points

Which of the following best describes JPA (Java Persistence API)?

Selected Answer: ☒ A specification for object-relational mapping (ORM) in Java

Answers: ☐ A framework for building REST APIs

☒ A specification for object-relational mapping (ORM) in Java

☐ A database management system

☐ A library for handling HTTP requests

Question 4

What does IoC (Inversion of Control) mean in the context of the Spring Framework?

Selected Answer: ☒ The framework manages the object lifecycle and dependencies

Answers: ☒ The framework manages the object lifecycle and dependencies

☐ Developers control the flow of the application manually

☐ Objects control their own lifecycle and dependencies

☐ The control of transaction management is handled entirely by the database layer.

Question 1

@BeforeAll methods can access instance variables in PER_METHOD lifecycle mode.

Selected Answer: ☒ False

Answers: ☐ True
☒ False

Question 2

In which scenario would using @TestInstance(Lifecycle.PER_METHOD) be more appropriate than PER_CLASS?

Selected Answer: ☒ When each test requires a fresh instance to avoid unintended state sharing.

Answers: ☐ When tests need to share state.
☐ When lifecycle methods need to be non-static.
☒ When each test requires a fresh instance to avoid unintended state sharing.
☐ When optimizing resource-heavy setups.

Question 3

What is the primary responsibility of the service layer in a Spring Boot application?

Selected Answer: ☒ To manage business logic.

Answers: ☐ To handle HTTP requests.
☐ To interact directly with the database.
☐ To render views.
☒ To manage business logic.

Question 4

What is the main purpose of the repository layer in a Spring Boot application?

Selected Answer: ☒ To provide a clean API for accessing and manipulating data.

Answers: ☐ To handle user interface logic.
☒ To provide a clean API for accessing and manipulating data.
☐ To define RESTful web services.
☐ To create a git repository.